

ENERGY & CONS EN

CHAPTERS 7 & 8

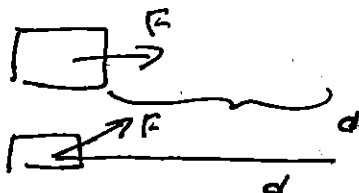
1/18

~~THE~~ ENERGY IS THE ABILITY TO DO WORK

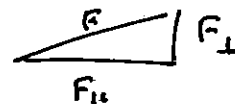
$$W = Fd$$

$$= F_{||} d$$

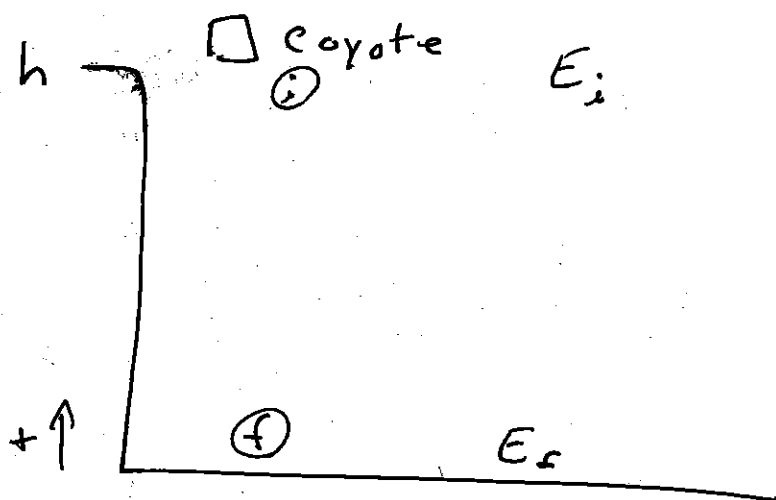
$$= \vec{F} \cdot \vec{d}$$



W is a scalar



$$[W] = [E] = \frac{\text{kg m}}{\text{s}^2} \text{ m} = \frac{\text{kg m}^2}{\text{s}^2} = \text{Joule}$$



Cons of E gives $E_i = E_f$

Kinetic $K = \frac{1}{2} m v^2$

Potential $U = mgh$

$$U_i + K_i = U_f + K_f$$

$$mgh_i + \frac{1}{2} m v_i^2 = mgh_f + \frac{1}{2} m v_f^2$$

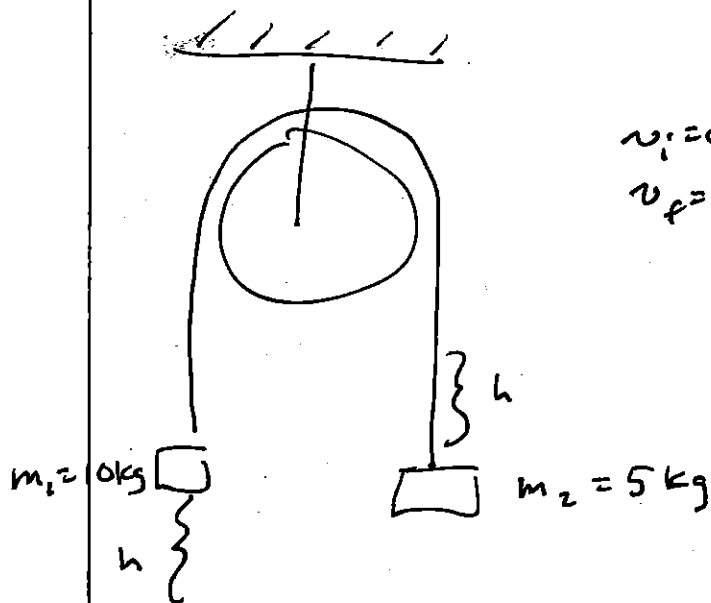
$$v_i = 0 \quad h_f = 0$$

$$mgh_i = \frac{1}{2} m v_f^2$$

$$v_f = \sqrt{2gh_i}$$

Atwood's Machine

2/18



$$v_i = 0$$

$$v_f = ? \text{ after } m_1 \text{ falls } h = 2 \text{ m}$$

$$\text{after } m_2 \text{ rises } 2 \text{ m}$$

m_2

$$E_i = E_f$$

$$h_i = 0$$

$$h_f = 2 \text{ m}$$

$$K_i + U_i = K_f + U_f$$

$$v_i = 0$$

$$v_f = ?$$

$$\frac{1}{2} m_1 v_{1i}^2 + \frac{1}{2} m_2 v_{2i}^2 + m_1 g h = \frac{1}{2} m_1 v_{1f}^2 + \frac{1}{2} m_2 v_{2f}^2 + m_2 g h$$

$$m_1 g h = \frac{1}{2} (m_1 + m_2) v_f^2 + m_2 g h$$

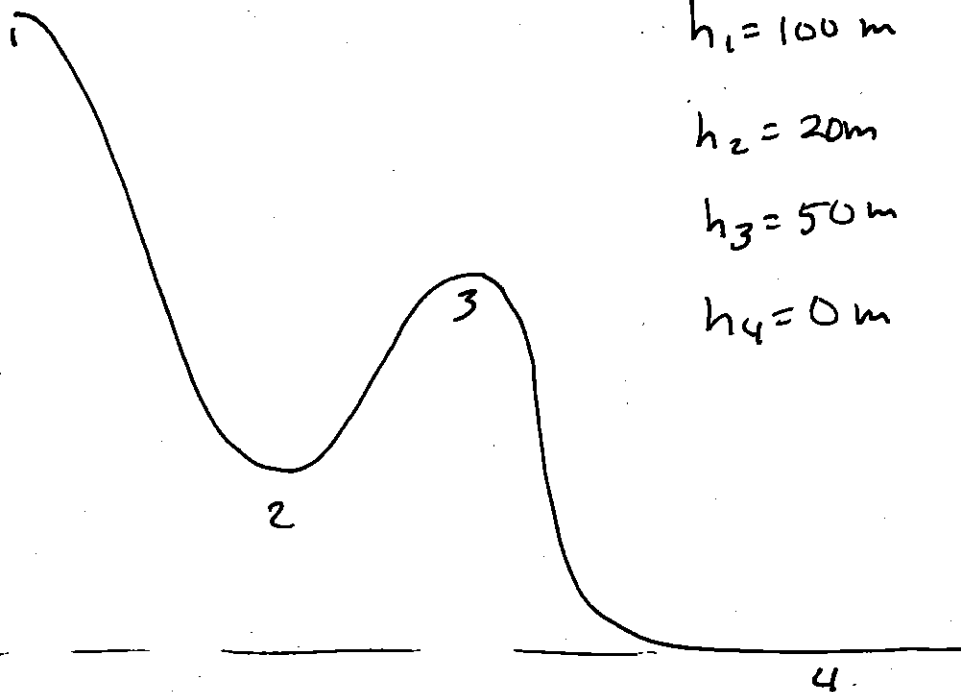
$$(m_1 - m_2) g h = \frac{1}{2} (m_1 + m_2) v_f^2$$

$$v_f = \sqrt{\frac{2 g h (m_1 - m_2)}{m_1 + m_2}} = \sqrt{\frac{2 \cdot 9.8 \cdot 2 \cdot (10 - 5)}{10 + 5}}$$

$$v_f = 3.61 \text{ m/s} \quad \text{up for } m_2 \quad \text{down for } m_1$$

Roller coaster

3/18



$$h_1 = 100 \text{ m} \quad v_1 = 0$$

$$h_2 = 20 \text{ m}$$

$$h_3 = 50 \text{ m}$$

$$h_4 = 0 \text{ m}$$

$$v_2 = ?$$

$$E_1 = E_2$$

$$K_1 + U_1 = K_2 + U_2$$

$$\frac{1}{2} m v_1^2 + m g h_1 = \frac{1}{2} m v_2^2 + m g h_2$$

$$m g (h_1 - h_2) = \frac{1}{2} m v_2^2$$

$$v_2 = \sqrt{2 g (h_1 - h_2)} = \sqrt{2 \cdot 9.8 (100 - 20)}$$

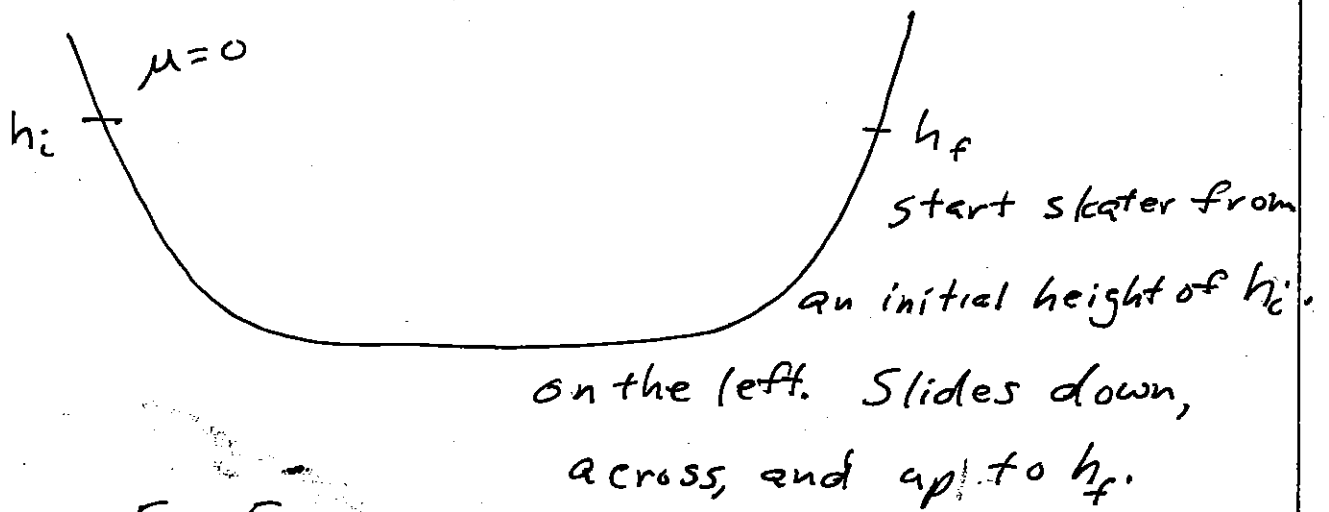
$$= 39.6 \text{ m/s}$$

$$v_3 = \sqrt{2 \cdot 9.8 (100 - 50)} = 31.3 \text{ m/s}$$

$$v_4 = \sqrt{2 \cdot 9.8 (100 - 0)} = 44.3 \text{ m/s}$$

CONSIDER A 'HALF PIPE'

4/18



$$E_i = E_f$$

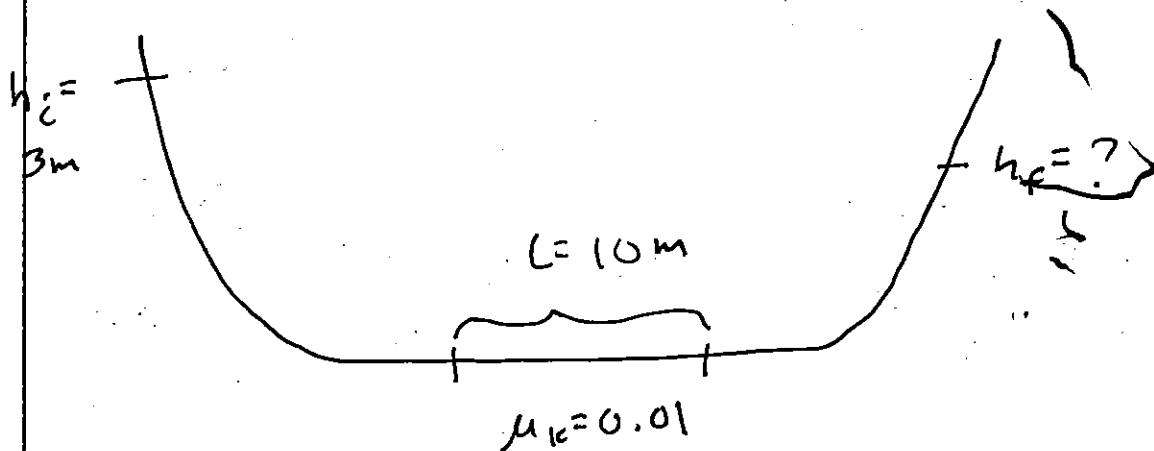
$$U_i + K_i = U_f + K_f$$

$0 \quad 0$

$$U_i = U_f$$

$$mgh_i = mgh_f \quad \boxed{h_i = h_f}$$

Now imagine it rains in the night, dust is washed down in a puddle which evaporates, leaving 10 m in a path w/ $\mu_k = 0.01$



Work-Energy Theorem

$$E_i = E_f + E_{\text{lost}}$$

lost E is due to friction 5/18

$$E_{\text{lost}} = F_f L$$

$$= \mu_k mg L$$

$$U_i = U_f + \mu_k mg L$$

$$mgh_i = mgh_f + \mu_k mg L$$

$$h_i = h_f + \mu_k L$$

$$h_f = h_i - \mu_k L = 3\text{m} - 0.01 \cdot 10\text{m} = 2.9\text{m}$$

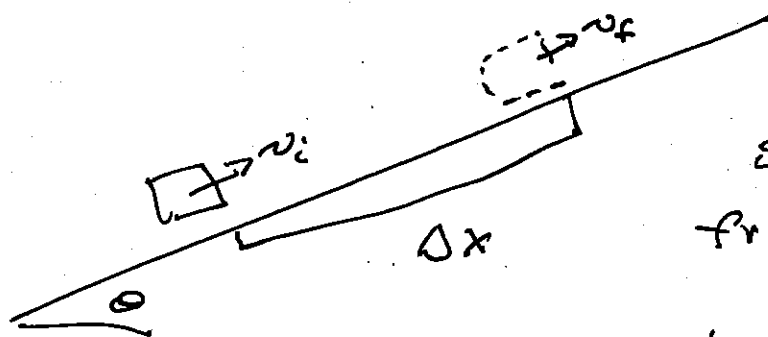
Energy was lost from the motion due to friction.

W-E Theorem in General

$$E_i + E_{\text{added}} = E_{\text{f}} + E_{\text{lost}}$$

E_{added} could be people pushing the skater as he dropped on the left.

6/18



sliding up a
frictionless incline
by cons of Energy

$$\theta = 10^\circ$$

$$\Delta x = 3 \text{ m}$$

$$v_i = 4 \text{ m/s}$$

how fast is it going at 3m?

$$E_i = E_f$$

$$U_i = 0$$

$$K_i = K_f + U_f$$

$$U_f = mgh$$

$$h = \Delta x \sin \theta$$

$$\frac{1}{2} m v_i^2 = \frac{1}{2} m v_f^2 + mg \Delta x \sin \theta$$

$$v_i^2 = v_f^2 + 2g \Delta x \sin \theta$$

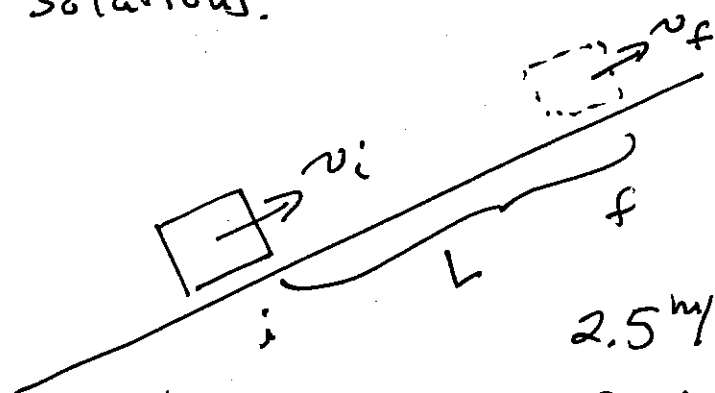
$$v_f = \sqrt{v_i^2 - 2g \Delta x \sin \theta}$$

$$v_f = \sqrt{4^2 - 2 \cdot 9.8 \cdot 3 \cdot 0.174}$$

$$= \sqrt{16 - 10.21}$$

$$= 2.41 \text{ m/s} \rightarrow$$

WET lets us revisit ugly old
force and kinematic problems for short
solutions.



$$\mu_k = 0.1$$

$$\theta = 30^\circ$$

A block 5kg slides
up a 30° incline at
 2.5 m/s . The coefficient of
kinetic friction is 0.1. When
the block has slid 0.5m, how
fast is it still moving?

$$E_i = E_f + E_{\text{lost}}$$

$$F_f = \mu_k mg \cos \theta$$

$$mgh_i + \frac{1}{2}mv_i^2 = mgh_f + \frac{1}{2}mv_f^2 + F_f L$$

$$h_i = 0 \quad h_f = L \sin \theta$$

$$\frac{1}{2}mv_i^2 = mgL \sin \theta + \frac{1}{2}mv_f^2 + \mu_k mg \cos \theta L$$

$$v_i^2 = 2gL \sin \theta + v_f^2 + 2\mu_k gL \cos \theta$$

$$v_f = \sqrt{v_i^2 - 2gL(\sin \theta + \mu_k \cos \theta)}$$

$$v_f = \sqrt{2.5^2 - 2 \cdot 9.8 \cdot 0.5 (0.5 + 0.1 \cdot 0.866)}$$

$$v_f = \sqrt{6.25 - 9.8(0.5866)} \\ \sqrt{6.25 - 5.75} \\ \sqrt{0.501}$$

8/18

$$v_f = 0.708 \text{ m/s}$$

$$\text{if } v_f = \sqrt{v_i^2 - 2gL(\sin\theta + \mu_k \cos\theta)} \\ = \sqrt{< 0}$$

it means block stops before traveling L.

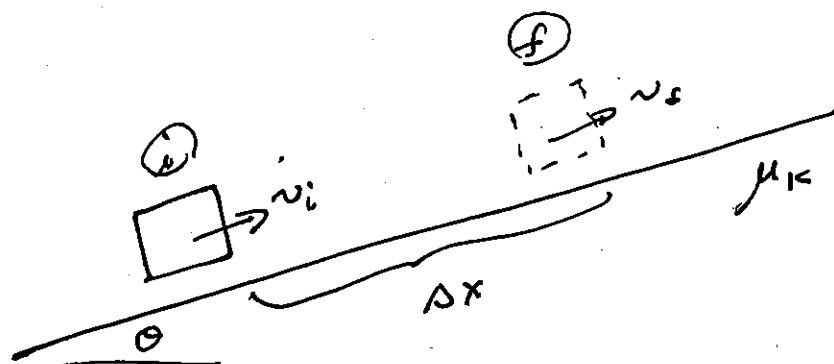
Drop back to find L

$$v_f^2 = v_i^2 - 2gL(\sin\theta + \mu_k \cos\theta)$$

$$L = \frac{v_i^2 - v_f^2}{-2g(\sin\theta + \mu_k \cos\theta)}$$

$$L = \frac{v_i^2 - v_f^2}{2g(\sin\theta + \mu_k \cos\theta)}$$

98/18



$$v_i = 3 \text{ m/s}$$

$$\theta = 16^\circ$$

$$\mu_k = 0.2$$

$$v_f = ? \text{ when } \Delta x = 2 \text{ m}$$

$$E_i = E_f + E_{\text{lost}}$$

$$\frac{1}{2} m v_i^2 = \frac{1}{2} m v_f^2 + m g \Delta x \sin \theta + \mu_k m g \cos \theta \Delta x$$

$$v_f^2 = v_i^2 - 2 g \Delta x (\mu_k \cos \theta + \sin \theta)$$

$$v_f = \sqrt{3^2 - 2 \cdot 9.8 \cdot 2 (0.2 \cdot 0.985 + 0.174)}$$

$$= \sqrt{9 - 14.54}$$

imaginary, so where does it stop?

$$v_f = 0$$

$$\Delta x = ?$$

$$0 = v_i^2 - 2 g \Delta x (\mu_k \cos \theta + \sin \theta)$$

$$\Delta x = \frac{v_i^2}{2 g (\mu_k \cos \theta + \sin \theta)} = \frac{9}{2 \cdot 9.8 \cdot 0.371}$$

$$\boxed{\Delta x = 1.24 \text{ m}} \quad \text{sliding up}$$

now sliding down

10
18

v_i $\leftarrow$

$$\mu_k = 0.2 \quad \theta = 10^\circ$$

$$v_i = 3 \text{ m/s}$$

$$\Delta x = 2 \text{ m}$$

$$mgh_i + \frac{1}{2} m v_i^2 = \frac{1}{2} m v_f^2 + \mu_k \cos \theta \, m g \Delta x$$

$$g \Delta x \sin \theta + \frac{1}{2} v_i^2 = \frac{1}{2} v_f^2 + \mu_k g \Delta x \cos \theta$$

$$v_f^2 = v_i^2 + 2 g \Delta x (\sin \theta - \mu_k \cos \theta)$$

$$v_f = \sqrt{3^2 + 2 \cdot 9.8 \cdot 2 (0.174 - 0.2 \cdot 0.985)}$$

$$= \sqrt{9 + 39.2 (-0.023)}$$

$$= \sqrt{8.0984}$$

$$\boxed{v_f = 2.85 \text{ m/s at } 2 \text{ m}}$$

$$\Delta x_{\max} = ? \quad v_f = 0$$

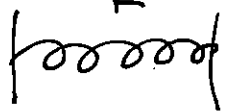
$$\Delta x = \frac{-v_i^2}{2g(\sin \theta - \mu_k \cos \theta)} = \frac{v_i^2}{2g(\mu_k \cos \theta - \sin \theta)}$$

$$= \frac{9}{2 \cdot 9.8 (0.2 \cdot 0.985 - 0.174)} = \boxed{19.96 \text{ m}}$$

sliding down

Hooke's Law Springs

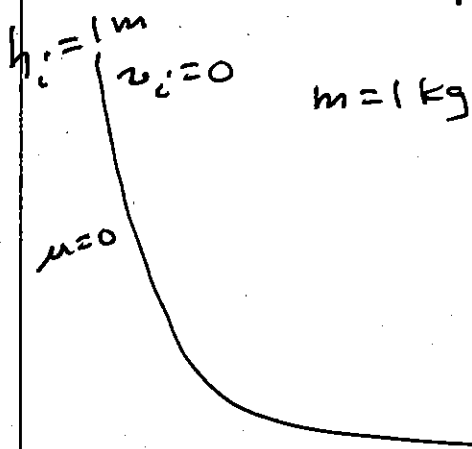
11/18

k
 Coils don't touch

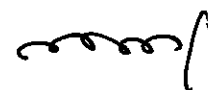
x_{eq}

$\leftarrow x \quad \rightarrow F = -kx \quad U = \frac{1}{2} kx^2$

$x \rightarrow \leftarrow F = -kx \quad U = \frac{1}{2} kx^2$



$k = 1000 N/m$



$x \rightarrow$
 $v_f = 0$
 $x = ?$

$E_i = E_f$

$mgh = \frac{1}{2} kx^2$

$x = \sqrt{\frac{2mgh}{k}} = \sqrt{\frac{2 \cdot 1kg \cdot 9.8m/s^2 \cdot 1m}{10^3 N/m}}$

$= 0.14m$

power

13/

18

$$[F] = N$$

$$[W] = \text{energy} = J$$

$$[P] = \frac{W}{t} = \frac{J}{s} = \text{Watts}$$

$$P = \frac{mgh}{t}$$

mg is weight

h is height

t is time

man 100 kg climbs 25 m stairs in

20 s

$$P = \frac{100 \text{ kg} \cdot 9.8 \frac{\text{m}}{\text{s}^2} \cdot 25 \text{ m}}{20 \text{ s}} = 1225 \text{ W} \quad (\text{hair dryer})$$

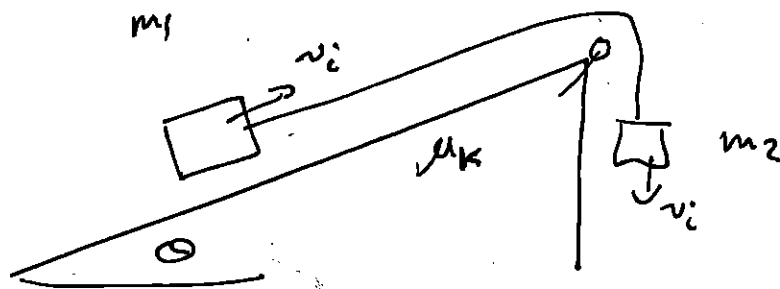
$$P = \frac{F d}{t}$$

$$= F v$$

Force at a speed v

power to move car!

15博
18



$$E_i = E_f + E_{lost}$$

$$\frac{1}{2}(m_1 + m_2)v_i^2 + m_2 g \Delta x =$$

$$\frac{1}{2}(m_1 + m_2)v_f^2 + m_1 g \Delta x \sin \theta + m_1 g \cos \theta \mu_k \Delta x$$

$$v_i^2 + \frac{2m_2 g \Delta x}{m_1 + m_2} = v_f^2 + \frac{2m_1 g \Delta x \sin \theta}{m_1 + m_2}$$

$$+ 2 \frac{m_1 g \cos \theta \mu_k \Delta x}{m_1 + m_2}$$

$$v_f^2 = v_i^2 + \frac{2g \Delta x}{m_1 + m_2} (m_2 - m_1 \sin \theta - \mu_k m_1 \cos \theta)$$

$$v_f = \sqrt{2^2 + \frac{2 \cdot 9.8 \cdot 0.5}{22} (2 - 20 \cdot 0.5 - 0.4 \cdot 20 \cdot 0.866)}$$

$$v_f = \sqrt{4 + 0.445 (2 - 10 - 6.928)}$$

$$v_f = \sqrt{4 - 6.64}$$

imaginary. Doesn't go 0.5 m.

$$m_1 = 20 \text{ kg}$$

$$m_2 = 2 \text{ kg}$$

$$\theta = 30^\circ$$

$$\mu_k = 0.4$$

$$v_i = 2 \text{ m/s}$$

$$\Delta x = 0.5 \text{ m}$$

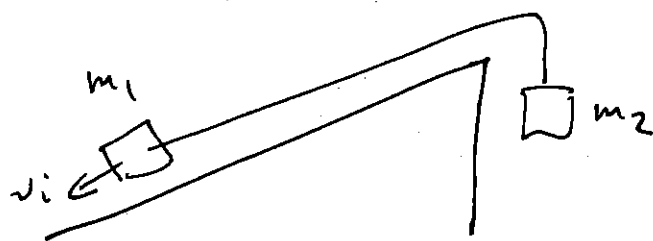
15/18

$$\Delta x = \frac{(m_1 + m_2)(v_f^2 - v_i^2)}{2g(m_2 - m_1 \sin \theta - \mu_k m_1 \cos \theta)}$$

$$\Delta x = \frac{22(0 - 2^2)}{2 \cdot 9.8(2 - 20 \cdot 0.5 - 0.4 \cdot 20 \cdot 0.866)}$$

$$\Delta x = \frac{-88}{19.6(2 - 10 - 6.928)}$$

$$\boxed{\Delta x = 0.30 \text{ m}} \quad \text{goes } 0.3 \text{ m up}$$



17/18

$$E_i = E_f + E_{\text{lost}}$$

$$\frac{1}{2}(m_1 + m_2)v_i^2 + m_1 g \Delta x \sin \theta = \frac{1}{2}(m_1 + m_2)v_f^2 + m_2 g \Delta x + \mu_k m_1 g \cos \theta \Delta x$$

$$(m_1 + m_2)v_f^2 = (m_1 + m_2)v_i^2 + 2g \Delta x (m_1 \sin \theta - m_2 - \mu_k m_1 \cos \theta)$$

$$v_f = \sqrt{v_i^2 + \frac{2g \Delta x}{m_1 + m_2} (m_1 \sin \theta - m_2 - \mu_k m_1 \cos \theta)}$$

$$v_f = \sqrt{2^2 + \frac{2 \cdot 9.8 \cdot 0.5}{22} (20 \cdot 0.5 - 2 - 0.4 \cdot 20 \cdot 0.866)}$$

$$= \sqrt{4 + 0.445 (10 - 2 - 6.928)}$$

$$= \sqrt{4 + 0.477}$$

$$= 2.116 \text{ m/s} \quad \text{it speeds up!}$$

what would slow it down?

if m_2 hanging were bigger
then m_1 would slow

18/18

$$m_1 = 20 \text{ kg} \quad m_2 = 10 \text{ kg}$$

$$v_f = \sqrt{2^2 + \frac{2 \cdot 9.8 \cdot 0.5}{30} (20 \cdot 0.5 - 10 - 0.4 \cdot 20 \cdot 0.866)}$$

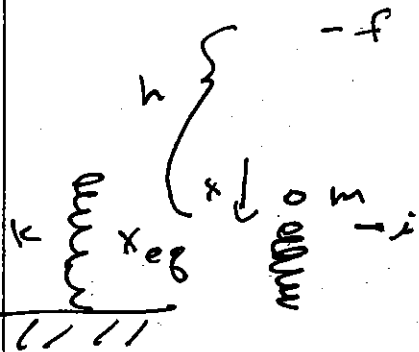
$$v_f = \sqrt{4 + 0.3267 (10 - 10 - 6.928)}$$

$$v_f = 1.32 \text{ m/s} \quad \text{after sliding down } 0.5 \text{ m.} \\ \text{to left}$$

or reduce θ

or increase μ_k

or reduce m_1



Compress a spring k by a length x . Place marble m on spring, release, How high does marble fly?

$$E_i = E_f$$

$$\frac{1}{2} kx^2 = mgh$$

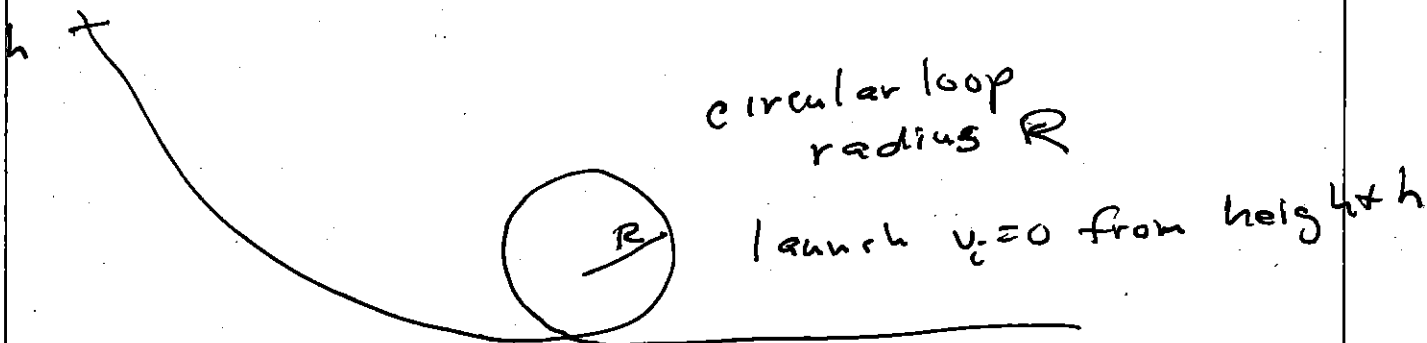
$$h = \frac{kx^2}{2mg}$$

$$k = 10^3 \text{ N/m} \quad x = 0.01 \text{ m}$$

$$m = 0.01 \text{ kg}$$

$$h = \frac{10^3 \text{ N/m} (0.01)^2 \text{ m}^2}{2 (0.01 \text{ kg}) (9.8 \text{ m/s}^2)} = 0.51 \text{ m}$$

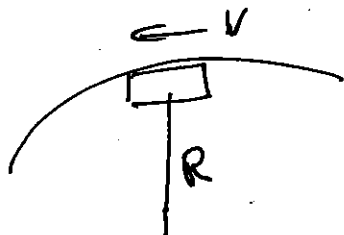
Roller Coaster Loop the loop



circular loop
radius R

launch $v_i = 0$ from height h

How high must the launch height be if car stays on the track (barely) at top?



$$\downarrow \frac{mv^2}{r}$$

$$N + mg = \frac{mv^2}{R}$$

$$N = \frac{mv^2}{R} - mg$$

barely means $N = 0$

$$\frac{mv^2}{R} = mg$$

$$v = \sqrt{gR}$$

v at top of loop not pressing into track or pulled from track

13/18

$$v = \sqrt{9.8 \cdot 10 \text{ m}^2/\text{s}^2} \quad R = 10 \text{ m}$$

$$v = 9.9 \text{ m/s} \quad \text{top}$$

~~OK, so what is N at bottom then?~~

$$h = ? \quad E_i = mgh$$

$$E_f = mg2R + \frac{1}{2}mv_{\text{top}}^2$$

$$mgh = mg2R + \frac{mgR}{2}$$

$$h = 2\frac{1}{2}R = \frac{5}{2}R = 25 \text{ m}$$

OK, so what is N at bottom?

$$E_i = E_f$$

$$E_i = mgh = E_f = \frac{1}{2}mv^2$$

$$v = \sqrt{2gh} \quad (\text{dropped coyote})$$



$$\uparrow \frac{mv^2}{R}$$

$$N - mg = \frac{mv^2}{R}$$

$$N = mg + \frac{m}{2}(2g \cdot \frac{5}{2}R)$$

$$N = 6mg$$

death &

unconsciousness